

Final model · Stage 3: the decision layer

From a probability to an action · validation split only (2015 H1) · run locally · test still sealed

Result: **APPROVE** if $PD \leq 22.5\%$ (64.7% of applicants, 11.9% default), **REVIEW** in between (9.8%), **DECLINE** if $PD \geq 27.5\%$ (25.5%, 40.1% default). Predicted expected loss is **0.94×** the real loss. All required checks pass.

A. How much is lost when a loan defaults?

Measured on the **66,928 defaulted training loans** (2012-08..2014-12), using repayment and recovery columns from the raw file. These are outcomes known only after a loan ends, so they were never model inputs. **EAD** (exposure at default) = principal still owed when the borrower stops paying; **LGD** (loss given default) = the part of it never recovered, after collection-agency fees.

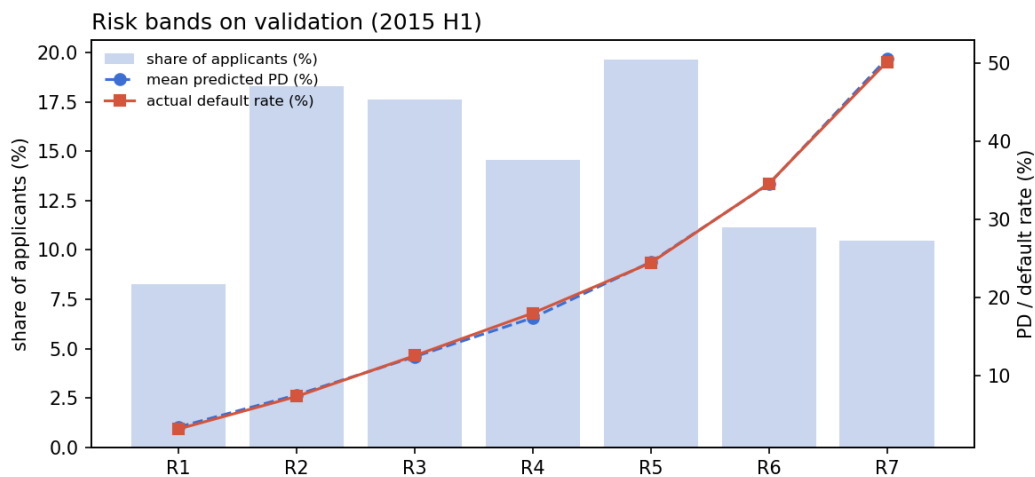
Term	Defaulted training loans	Share still owed at default (EAD share)	Loss given default (LGD)	Loss per \$ funded (EAD share × LGD)
36 months	37,949	57.3%	88.7%	50.8%
60 months	28,979	70.5%	88.8%	62.7%
all	66,928	64.7%	88.8%	57.4%

Expected loss (EL) = $PD \times \text{loan amount} \times \text{EAD share} \times \text{LGD}$. Example: a \$10,000 loan with PD 20% → $0.20 \times 10,000 \times 0.57 \times 0.89 = \$1,015$ for 36 months, \$1,253 for 60 months (more principal is still owed when a longer loan defaults). Recoveries are small: a quarter of defaults recovered nothing.

B. Risk bands

Seven fixed PD ranges give every applicant a short label, like a grade. The requirement (fixed beforehand) is that the actual default rate rises from band to band. It does, and each band's mean PD matches its actual default rate.

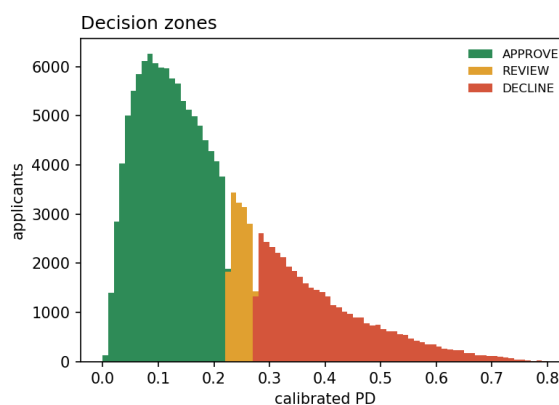
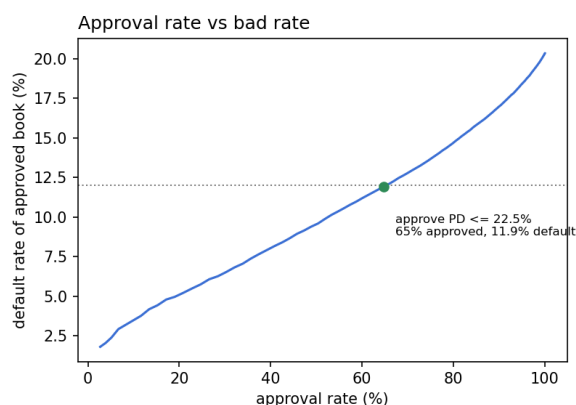
Band	PD range	Share	Mean PD	Actual default	Loss per \$ funded
R1 very low risk	0% – 5%	8.3%	3.5%	3.2%	1.4%
R2 low risk	5% – 10%	18.3%	7.6%	7.4%	3.4%
R3 moderate risk	10% – 15%	17.6%	12.4%	12.6%	6.5%
R4 elevated risk	15% – 20%	14.6%	17.4%	18.0%	10.0%
R5 high risk	20% – 30%	19.6%	24.5%	24.4%	14.6%
R6 very high risk	30% – 40%	11.1%	34.5%	34.6%	22.2%
R7 severe risk	$\geq 40\%$	10.5%	50.5%	50.1%	34.5%



C. Approve / review / decline

The thresholds come from two business targets written in the plan before any result was seen. They were searched on a 0.5-point PD grid using out-of-fold calibrated PDs: **APPROVE** the largest group whose actual default rate is $\leq 12\%$; **DECLINE** the riskiest group whose default rate is $\geq 40\%$ (two in five default); send the rest to a person for **REVIEW**.

Decision	Rule	Share	Mean PD	Actual default	Expected loss	Real loss
APPROVE	PD $\leq 22.5\%$	64.7%	11.8%	11.9%	\$91.0M	\$90.7M
REVIEW	in between	9.8%	24.9%	24.6%	\$33.0M	\$34.2M
DECLINE	PD $\geq 27.5\%$	25.5%	40.2%	40.1%	\$172.9M	\$190.4M



D. Does expected loss match reality?

Group	Expected	Real	Ratio
all	\$296.8M	\$315.3M	0.941
term 36 months	\$117.6M	\$117.9M	0.997
term 60 months	\$179.3M	\$197.4M	0.908

Required: within $\pm 10\%$ overall (by decision group: see table C). Recoveries on 2015 loans were still coming in when the data ends (2018), so the real loss is, if anything, slightly overstated.

E. What it returns (a declined applicant)

```
{
  "probability_of_default": 0.383,
  "risk_band": "R6",
  "decision": "DECLINE",
  "loan_amount": 24000, "term_months": 60,
  "expected_loss_usd": 5759.47,
  "reasons": [
    "Loan term: 60 months",
    "Loan amount: $24,000",
    "Loan title (historical default rate): 21.7%",
    "Months since the most recent credit inquiry: 0 months",
    "strengths": ["Annual income: $150,000", "Debt-to-income ratio: 3.3%"]
  ]
}
```

Actual outcome: defaulted. Reasons come from the Stage 2 SHAP reason codes.

F. Business impact on validation (losses only)

Policy on the 162,745 validation loans	Approved	Default rate	Real loss	Loss per \$ funded	Loss avoided
approve everyone (what LendingClub funded)	100.0%	20.4%	\$315.3M	13.2%	—
engine, strict (review = decline)	64.7%	11.9%	\$90.7M	6.2%	\$224.6M
engine, lenient (review = approve)	74.5%	13.6%	\$124.9M	7.4%	\$190.4M

What we learned

- The strict policy cuts real losses from \$315.3M to \$90.7M (6.2% instead of 13.2% of every dollar lent), but it also turns away 36,849 borrowers who repaid. Their lost interest is not counted here because the model does not set prices; a real lender would weigh both.
- The approve threshold (22.5%) falls inside band R5 (20–30%), so R5 is split across all three decisions. Bands describe risk; thresholds are business choices. Both live in one editable file, *decision_policy.json*.
- 60-month loans: expected loss is $0.91 \times$ the real loss (36-month: $1.00 \times$). This matches the Stage 1 finding that 60-month PDs run about 1 point low; Stage 5 (test) will show whether a term-specific correction is needed.

Next: Stage 4, one scoring package (raw application $\rightarrow$ preprocessing $\rightarrow$ model $\rightarrow$ calibration $\rightarrow$ this decision layer $\rightarrow$ JSON).